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Preliminary Design Review - Electrical

INNEX-1 Rover - Innova UoL

University of Leicester, UK Lunabotics Team

Team Name	Innova UoL
Project Name	INNEX-1 Rover
Sub-Team	Electrical
Document Title	PDR - Electrical
University	University of Leicester
Department	School of Engineering
Date of Submission	31/01/2026
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Version	1.3

Summary

This report details the electrical design for the INNEX-1 Rover, prioritizing reliable autonomy and safety compliance. We selected the NVIDIA Jetson Orin Nano over the Raspberry Pi 5 because our 3D SLAM workload requires a dedicated GPU to run in real-time without latency bottlenecks, which is a margin of safety the Pi couldn't offer. To prevent high-current motor stalls from crashing the computer, we implemented a Dual-Source Power Architecture that physically isolates the 22.2V actuation rail from the 14.8V compute rail. Our safety system follows a moving part off and data part on strategy, using a E-Stop and master switch to physically sever motor power during an E-Stop event while keeping the Jetson alive to preserve critical telemetry and logs for post-failure analysis.

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1 Compute platform selection

The selection of the primary computing unit was driven by a trade-off analysis focusing on three critical performance metrics: AI inference capability, power/thermal efficiency, and I/O connectivity.

While the Raspberry Pi 5 offers a strong general-purpose ecosystem, the NVIDIA Jetson Orin Nano was selected as the rover's central processor due to its specialized architecture. Unlike standard microcontrollers (Arduino Giga) or general-purpose Single Board Computers (Raspberry Pi 5), the Orin Nano integrates a dedicated GPU capable of parallel processing. This computational headroom is critical for the rover's autonomy, enabling real-time sensor fusion and SLAM without saturating the CPU.

The comparative analysis below details the technical specifications of the platforms and highlights the architectural advantages of the selected Jetson system.

	Orin Nano	Pi 5	Arduino Giga
Price	£186-250	£90-130	£50-153
CPU	6 core arm cortex A784E 64bit	2.4GHz quad-core 64-bit ArmCortex-A76	480 MHz Dual-core Arm Cortex-M7
GPU/Accelerator	GPU - Nvidia Ampere gpu 1024 CUDA cores + 32 tensor cores	Accelerator - Broadcom VideoCore 7 800MHz	Accelerator - Chrom-ART Accelerator (DMA2D), LTDC
RAM	8GB LPDDR5, 128-bit wide, 68Gb/s bandwidth	4GB or 8GB LPDDR4X	1 MB of internal SRAM, 2 MB of onboard Flash, 8 MB of external SDRAM.
Power	5-10W	3-5W	5W
Storage	microSD + M.2 NVMe	high-speed NVMe SSDs via a PCIe connector and an M.2 HAT/adaptor	2MB internal flash, 1MB SRAM + microSD
Ports and connectivity	4x USB 3.2 Gen 2 Type-A ports, a data-only USB-C port, a Gigabit Ethernet port, and	Gigabit Ethernet (with PoE+ support), dual-band 802.11ac Wi-Fi, Bluetooth 5.0/BLE, and dual 4K micro-HDMI ports	Wi-Fi 4, Bluetooth 5.1, USB-C host/device, microSD, multiple UART/I ² C/SPI/CAN interfaces, and

	a DisplayPort 1.2 video output		camera/display support (MIPI DSI/CSI + LTDC)
Camera Limitation	can capture multiple camera streams, but if you need to encode (compress) and stream or record them — especially simultaneously — you are likely to saturate CPU cores and may struggle to maintain stable frame rates, especially at high resolution / high framerate	two 4-lane MIPI CSI-2 connectors, enabling dual-camera support, yet it faces limitations with high-resolution HDR (max 2304×1296), potential 2GB memory allocation bottlenecks for imaging pipelines, and software-dependent support for complex multi-camera setups. While supporting higher data rates, it retains standard 1080p.	interface with cameras via MIPI CSI , but it has significant limitations compared to single-board computers like the Raspberry Pi. It cannot handle high-resolution or high-frame-rate video , is limited by available RAM for image buffers , and lacks a GPU for hardware-accelerated image processing, so most camera processing must be done on the CPU cores . It's suitable for low-resolution snapshots, simple image capture, or basic computer vision , but not for streaming HD video, complex image processing, or real-time 3D vision.
General Limitations	- No hardware encoder - USB bandwidth	- Weaker CPU - No GPU - USB bandwidth - ROS2 support is worse - Thermal limitations	- No GPU / 3D acceleration - Limited RAM - Limited flash - 3.3 V logic only - No Linux / OS support

	- Thermal limitations under AI - Memory bandwidth	- No ROS2 friendly hardware encoder	- Ethernet requires external PHY
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1.1.1 Why We Selected the Jetson Orin Nano

Jetson Orin Nano vs. Raspberry Pi 5: While the Raspberry Pi 5 is an excellent general-purpose computer for hobbyist projects, it lacks the specialized hardware required for autonomy. The Jetson Orin Nano features a dedicated 1024-core NVIDIA Ampere GPU and 40 TOPS of AI performance, which are critical for processing our heavy sensor load (Unitree L1 LiDAR + OAK-D Cameras).

The Pi 5 relies heavily on its CPU for processing 3D point clouds and depth data, which would lead to high latency and laggy mapping (SLAM).

The Jetson offloads these tasks to its GPU, ensuring smooth, real-time navigation and leaving the CPU free for mission logic. For a competition rover, the Pi 5 is a risk; the Jetson is a guarantee.

Jetson Orin Nano vs. Arduino Giga: The Arduino Giga and the Jetson operate in completely different domains. The Arduino is a microcontroller designed for real-time actuator control, whereas the Jetson is a computer designed for high-level intelligence.

We cannot run SLAM or computer vision on an Arduino.

The Jetson handles the Mapping, Path Planning and sends high-level commands to our motor drivers, creating a distributed architecture.

Overall Verdict: The NVIDIA Jetson Orin Nano is the clear winner for this project. While the Raspberry Pi 5 is cheaper, it offers no safety margin for the computational demands of 3D LiDAR and stereo vision. The Jetson provides the GPU acceleration necessary for modern ROS2 robotics, ensuring our rover can map, plan, and react in real-time without computational bottlenecks. It transforms our machine from a remote-controlled car into a truly autonomous robotic system.

Hardware | Computation

Compute Platform Comparison

Jetson Orin Nano



- Price: £186-250
- CPU: 6-core ARM Cortex-A78AE 64-bit
- GPU: NVIDIA Ampere (1024 CUDA + 32 Tensor Cores)
- RAM: 8GB LPDDR5
- Power: 5-10W
- Storage: microSD + M.2 NVMe
- Storage: microSD + M.2 NVMe
- Ports: 4x USB 3.2 Gen 2, USB-C (Data), GbE, DP 1.2
- Limitations: No hardware encoder; CPU bottleneck for high-res recording/streaming; Thermal limits

Raspberry Pi 5



- Price: £90-130
- CPU: 2.4GHz Quad-Core ARM Cortex-A76 64-bit
- Accelerator: Broadcom VideoCore 7 800MHz
- RAM: 4GB or 8GB LPDDR4X
- Power: 3-5W
- Storage: microSD + M.2 NVMe (via HAT)
- Ports: GbE (PoE+), WiFi 5, BT 5.0, Dual 4K micro-HDMI
- Limitations: Weaker CPU, No GPU, USB bandwidth, Limited HDR, 2GB imaging pipeline bottleneck

Arduino Giga



- Price: £50-153
- CPU: 480 MHz Dual-Core ARM Cortex-M7
- Accelerator: Chrom-ART (DMA2D), LTDC
- RAM: 1MB SRAM, 2MB Flash, 8MB External SDRAM
- Power: 5W
- Storage: 2MB Flash, 1MB SRAM, microSD
- Ports: WiFi 4, BT 5.1, USB-C, multiple interfaces, camera/display support
- Limitations: No GPU/3D acceleration, Limited RAM/Flash, No Linux OS, 3.3V logic, External PHY for Ethernet. Suitable for basic CV, not HD video or 3D

2 Power architecture and E-Stop

The rover employs a Dual-Source Power Architecture designed to achieve complete physical isolation between the high-current actuation domain and the noise-sensitive computational domain. By utilizing two independent power sources, the system eliminates the risk of inductive voltage spikes, back-EMF, or voltage drops caused by motor stall currents, which could otherwise trigger brownouts or resets in the primary SBC. This topology ensures that the robotic autonomy stack remains stable even during high-load excavation or climbing manoeuvres where current draw peaks significantly.

2.1 Actuation Domain (High-Power)

The actuation subsystem is energized by a high capacity 6S LiPo battery, providing a nominal 22.2V to the drivetrain and excavation motors. Given the potential for high current draw during operation, this circuit is protected by a 60A Master Fuse located immediately after the battery terminals to prevent over-current events.

Safety is enforced via a hardware Emergency Stop (E-Stop) mechanism placed in series between the fuse and the power distribution point. Upon activation, this switch physically cuts the 22.2V supply to the breaker board, instantly de-energizing all motor drivers and halting mechanical motion without reliance on software logic.

The breaker board distributes the unregulated 12V rail to the individual motor controllers, which are rated for up to 43A to handle the thermal and current demands of the drive motors.

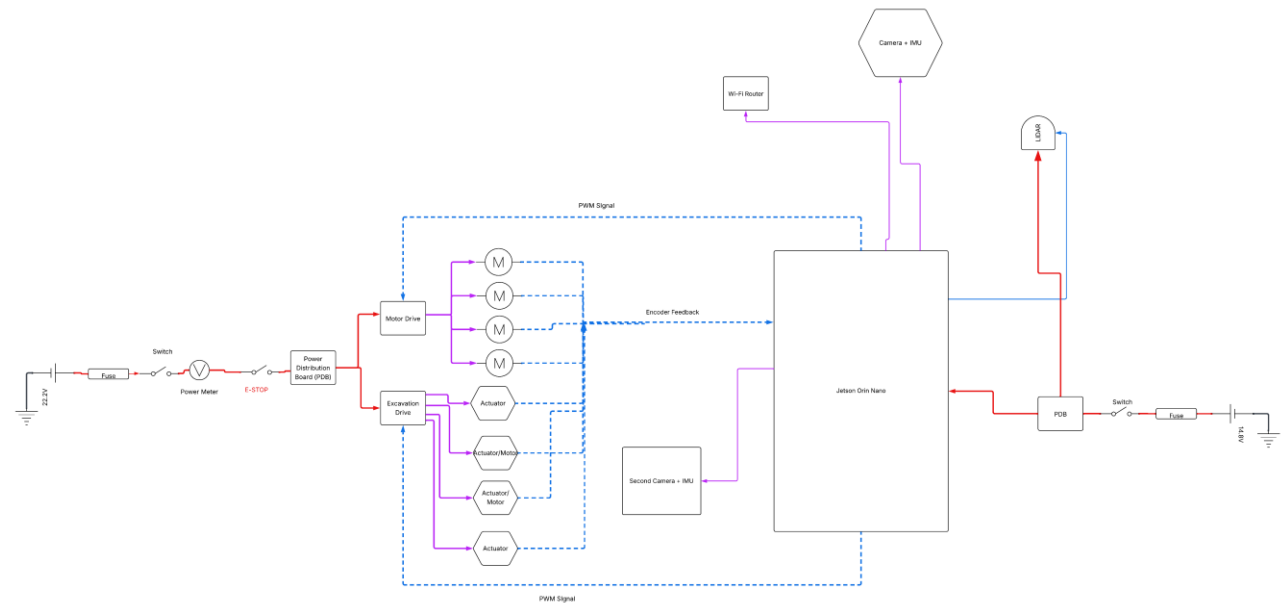
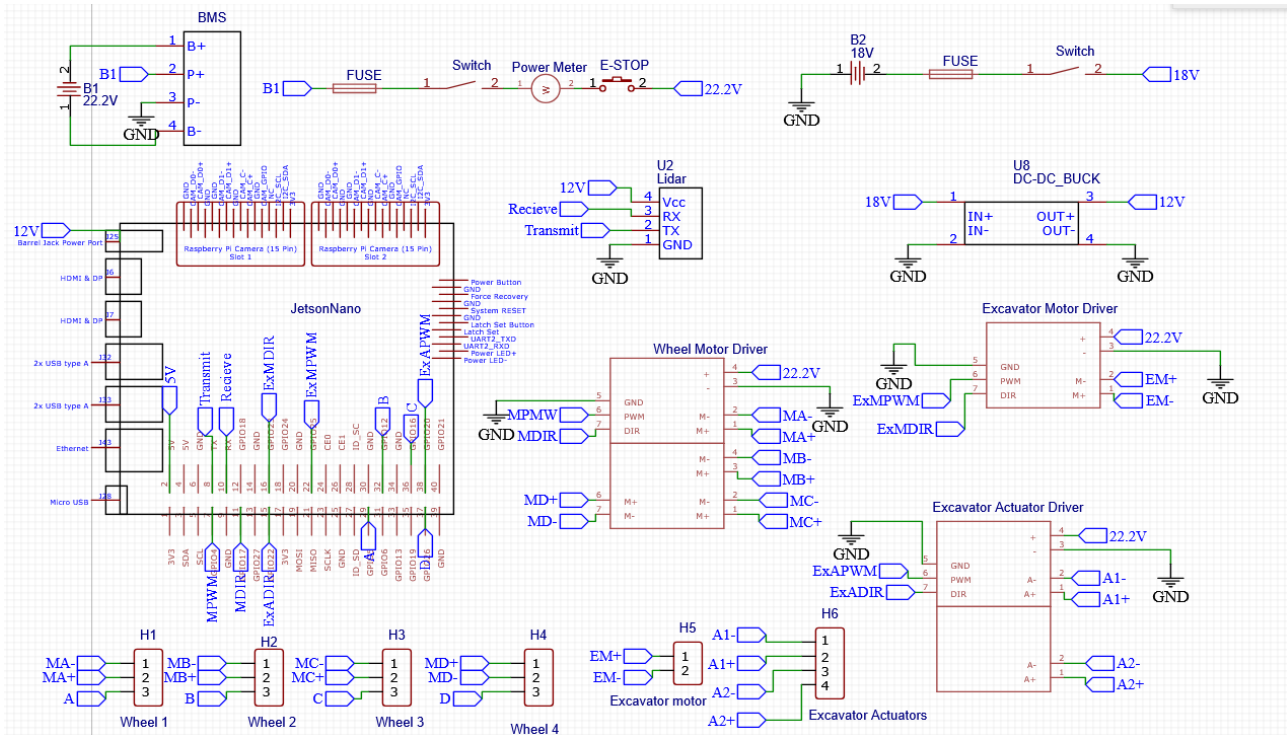
2.2 Computation Domain (Low-Power)

Parallel to the actuation circuit, the computational domain operates on a separate 4S LiPo battery supplying a nominal 14.8V. This power rail feeds the NVIDIA Jetson Orin Nano, the Unitree L1 LiDAR, and the peripheral vision systems. A defining feature of this architecture is its independence from the E-Stop safety loop, when the emergency stop is triggered to cut motor power, the compute domain remains fully energized. This ensures that system telemetry, sensor logs, and video feeds are preserved for analysis, allowing the team to diagnose the cause of a fault even after the mechanical systems have been deactivated.

2.3 Signal Interfacing and Grounding

To maintain signal integrity between the isolated power domains, the system utilizes a Common Ground architecture. While the positive voltage rails (22.2V and 14.8V) remain strictly separated to prevent cross-contamination, the negative terminals of both battery systems are tied to a single reference point. This common ground is essential for the 3.3V/5V logic signals sent from the Jetson to the motor drivers, ensuring that control signals are interpreted accurately relative to a stable baseline.

2.4 Schematics

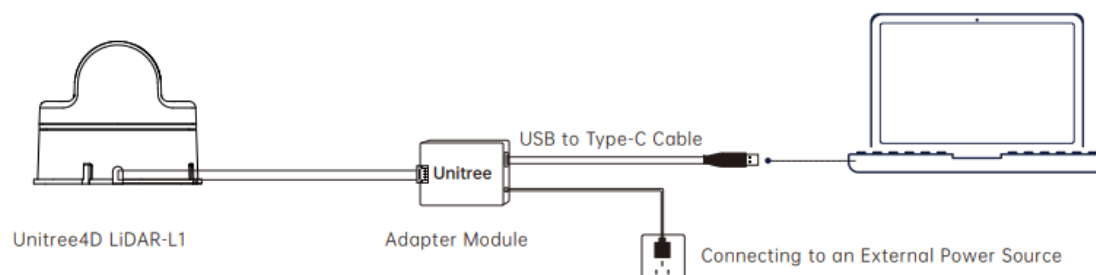


3 Sensor suite and communication budget

3.1 Sensor Suite Definition

Sensor	Role	Model	Interface
Primary Lidar	3D Mapping & Obstacle Avoidance	Unitree L1 (360° Omni)	USB/
Primary Vision	Object Detection & Depth	OAK-D Pro (Stereo + RGB)	USB 3.0
Secondary Vision	Reverse Direction Vision	OAK-D Camera	USB 3.0
IMU	Odometry / Orientation	Internal 6-Axis (Jetson/OAK)	SPI/I2C

3.1.1 Lidar



PIN NO.	Signal	Attribute	Description	Cable Color	Function
1	VCC	Power Supply	DC 12V power supply	Red	Power cable
2	GND	Ground	Ground	White	Power cable
3	RX	Input 3.3V	Ranging core serial port input	Green	Functional cable
4	TX	Output 3.3V	Ranging core serial port output	Black	Functional cable

3.1.2 Cameras and IMU

SYMBOL	RATINGS	MIN	TYP	MAX	UNIT
V_{BUS}	VBUS input supply voltage		5V	5.25	V
P	Power consumption requirement	4	6	7.5	W
P_{IDLE}	VBUS idle power draw (Myriad X booted)		2.5		W
T_A	Ambient operating temperature			50	°C

OAK-D Pro integrates a 9-Axis (Acceleration, Gyroscope, and magnetometer) BNO086 inertial measurement unit. MotionEngine 9-Axis and 6-Axis Sensor Fusion provide raw, calibrated sensor orientation data for more accurate heading and orientation.

Due to IR light sources the consumption of OAK-D Pro can be higher and will fall out of the USB2 maximum specified range of 900mA. If the host is capable of delivering only 900mA current over a single USB port then Y-adapter provides an option to have a separate power supply connected to the same USB device along with the data connection from the host. This allows that OAK-D Pro can still be run on devices such as Raspberry Pi4 and similar hosts with lower power capabilities on USB ports.

3.2 Bandwidth Budget

Bandwidth Considerations:

No#	Component name	Bandwidth Details
1	LiDAR	2.07 Mbps (without IMU data)
2	Camera	200Kbps (telemetry data only, main data is more than 1 Gbps)
3	IMU	IMU 1 takes between 0.06 to 0.05 Mbps (the lower end is with telemetry data, whilst the upper end is with 800 Hz)
16	Dual-Band Router	Total is about 2.32 Mbps on the upper end

$$Bandwidth = Sampling\ rate \cdot PointDataSize$$

General Considerations-

No#	Component name	Notes
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2	Camera	Use strong compression for low frame rate to remain under 4000 Kps average. Send depth slices and do onboard processing.
3	IMU	Document disabling of magnetometer
4	Main Battery	May need BMS
8	Voltage Regulator	May need LDO due to electrical noise
11	Microcontroller	Jetson real time processing so raw footage from cameras not needed, meaning can help use with bandwidth
12	Motor Controllers	4 for Drivetrain, 1 for excavation
14	Actuators	4 for Excavation and Deposition

4 Part list

Component name	Link to model	Quantity	Price:	Notes	Subsystem
LiDAR	Unitree 4D LiDAR L1	1	£182.82	Mapping and obstacle detection. Suggestion: mount with dust protection and could add	Sensors

				lens covers.	
Camera (RGB-D camera)	https://shop.luxonis.com/products/oak-d?srsltid=AfmBOor1hYN9651POX5riPHfE4wyqtrWrdUAFqG75CzTFxjDw_lj3eQ2	1	£250	Use strong compression or low frame rate to remain under 4000Kbps average. Send depth slices and do onboard processing.	Sensors

Main battery: 22.2V 24V	Tattu 10,000mAh 22.2v 30C 6S1P Lipo (EC5) – The Bionic Eye	1	First link: £196 Second link: £110.66	May need BMS	Power system
Secondary battery: 18V around 40C discharge (for Jetson) or a 11.1V Li-Po + boost converter	HRB 4s 14.8v 3000Mah 60C RC LiPo Battery with T Plug for Helicopter,E-flite Rare Bear, Fury 15 DF SIGP 3S Lipo Battery 11.1V 5600mAh 70C RC Battery with XT60 Connector for RC Vehicles Car Truck Tank Boat Racing Hobby: Amazon.co.uk: Toys & Games https://www.3dxr.co.uk/electronics-c78/power-management-c91/voltage-regulators-beecs-c101/hobbywing-5a-ubec-5v-6v-7-4v-2-8s-v2-air-p5267	1	First link: £30.52 Second link: £37.99 Third link: £10.99	The second link is a 11.1V LiPo battery and a step-down voltage converter.	Power system
Motor controllers	13Amp 6V-30V DC Motor Driver Sabertooth Dual 2x32A 6V-24V Regenerative	5	First link: £15 Second link: £28	Might need to investigate other drivers	Controls System

	Motor Driver - RobotShop (for the drive motors) RoboClaw 2x15A, 6-34VDC Regenerative Motor Controller - RobotShop (for excavation)		Third link: £114.14(estimated shipping - £13.78) Fourth link: £92.26(estimated shipping - £13.78)	as the DC motors and excavation motor have high torque. The third link is quite expensive, but it provides the continuous current that the chosen motors pull.	
Drivetrain Motor	GR-WM4-ENC 24V DC 37rpm Worm Gear Motor with Encoder - Gimson Robotics	4	£55		Power system
Excavation Motor	60MM Brushed Planetary Gear DC Motor 12V 24V CW/CCW High Power High Torque 60PG60S Shaft 12mm Keyway Motor - AliExpress 13	1	£59.99		Power system

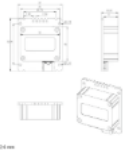
Onboard computer/microcontroller. Currently choosing between a Jetson nano or a Raspberry Pi5	Raspberry Pi 5 - The Pi Hut 945-13766-0005-000 NVIDIA Jetson Orin Nano Super Developer Kit Development Kit RS	1	£139.20 for the Pi5 (This is with the highest memory 16GB) £259.01 For the Jetson Orin Nano Super	Jetson does provide on board real-time processing so raw footage isn't required from cameras which can help use with bandwidth.	Controls system
Voltage regulators (Linear) Buck converter	https://uk.rs-online.com/web/p/voltage-regulators/2520795 8 Pack Mini MP1584EN DC-DC Buck Converter Adjustable Power Supply Module 24V to 12V 9V 5V 3V: Amazon.co.uk: Business, Industry & Science	2	£9.29 £10	May need an LDO (Low dropout regulator) because it has low noise therefore no interference with our	Power system

				sensors .	
Dual-band router (2.4 GHz + 5 GHz)	TP-Link Archer AX55 WiFi 6 Router, AX3000 Mbps wifi Booster, Dual Band Gigabit Ports, Internet Booster, Wi-Fi Extender, Ultra-Low Latency, 160MHZ Bandwidth, EasyMesh Compatible, Works with Alexa : Amazon.co.uk: Electronics & Photo	1		Must disable 2.4GHz and set the 2.4GHz channel to 20MHz.	Communications
Distribution board	WUPP Boat Fuse Block, Waterproof Fuse Panel with LED Warning Indicator Damp-Proof Cover - 12 Circuits with Negative Bus Fuse Box Holder for Car Marine RV Truck DC 12-24V, Fuses Included : Amazon.co.uk: Business, Industry & Science	1	£36.19		Power system
BMS (Battery Management System)	CIJIAOPOYU Lifepo4 Lithium BMS JKBMS 4S 5S 6S 7S 8S 17S 20S 24S 12V 24V 48V Active BMS 40A 60A 100A 200A 300A(JK-BD4A24S10P) : Amazon.co.uk: Business, Industry & Science (for main battery) iHaospace 5S Lithium Battery Protection	1 for the main battery and another for the secondary battery	First link: £99.50 Second link: £7		

	Circuit Board, 18V 21V 30A BMS, Motherboard Type, with Cable : Amazon.co.uk: Computers & Accessories (for the secondary battery)				
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4.1 New Components List

N o #	Com pone nt name	Q T Y .	Pri ce	Notes	Sub- syste m	Compa nies/Su ppliers	Schematic links
1	LiDAR	1	£48.71		Sensors	RS components	Not available as product is out of stock.

2	Cam era	1	£2 50		Sens ors	Luxonis	OAK-D – Luxonis
3	IMU	1	£5 2. 62	Docu ment disabl ing of magn etom eter.	Sens ors	RobotS hop new link: SYD Dynami cs Transdu cerM 9- Axis AHRS / IMU w/ Dual- Port Commu nication (TM171) - RobotS hop	 Dimensions are also on the website for a clearer view.
4	Main batte ry	1	£1 96		Powe r Syste ms	Tattu	tattu-16000mah-6s1p-15c-lipo-battery.pdf
5	Seco ndary batte ry	1	£3 7. 99		Powe r Syste ms	Amazon New link for a similar battery - HRB 3S Lipo Battery 11.1V 6000mA h 50C RC Battery	N/A Dimensions: 155 x 48 x 27 mm (0 - 3mm difference) Weight: 419g ± 2g

						With XT60 plug for DJI F450 Trex-500 RC Heli, RC Traxxas Car/Truck, RC Buggy, RC Boat and Drone: Amazon.co.uk: Toys & Games	
6	Drive train motor	4	£55		Power Systems	Gimson Robotics	GR-WM4-ENC_Summary-Specifications_05.09.23.pdf
7	Excavation motor	1	£59.99		Power Systems	AliExpress	On website
8	Voltage regulator	2	£10	May need LDO due to electrical noise.	Power Systems	Amazon	monolithicpower.com/en/documentview/productdocument/index/version/2/document_type/Datasheet/lang/en/sku/MP1584EN-LF-Z/document_id/204/
9	Power Distribution	1	£36.19		Power Systems	Amazon WUPP Boat Fuse Block,	Official documents are not released Product dimensions – 13.82 x 8.5 x 3.65cm ,

	Boar d					Waterproof Fuse Panel with LED Warning Indicator Damp-Proof Cover - 12 Circuits with Negative Bus Fuse Box Holder for Car Marine RV Truck DC 12-24V, Fuses Included : Amazon.co.uk: Business, Industry & Science	
1 1	Micro contr oller	1	£1 39 .2 0 or £2 59	Optio ns betwe en Jetso n and Rasp	Contr ol Syste ms	Raspberry Pi5 Jetson orin nano	NVIDIA Jetson Orin Nano Super Developer Kit NVIDIA

			.0 1	berry Pi5.			
1 2	Moto r contr ollers	5	£1 14 .1 4 an d 1 £9 2. 26	4 for drivet rain, 1 for excav ation.	Contr ol Syste ms	Saberto oth RoboCl aw	rb-dim-47-datasheet.pdf roboclaw-datasheet-2x15A.pdf
1 3	Dual- band route r	1	£5 1. 75		Com muni cation Syste m	Slate Plus router	a1300 datasheet EN 20230602
1 4	Cam era	1	\$4 99		Sens or	Gemini 435Le – Orbbec	Dimensions are on the website.

5 Interfaces are defined and consistent

5.1 Electrical Interfaces

Rail ID	Voltage	Source	Connector	Amperage Limit	Parts
P-HV (High Voltage)	22.2 VDC	6S LiPo Battery	XT60 (Yellow)	60A	Drive Motors (x4), Excavation motor, Excavation Actuators (x4)
P-LV (Low Voltage)	14.8 VDC	4S LiPo Battery	XT60 (Yellow)	7A	Jetson Orin Nano, Compute Domain
P-12V	12.0 VDC	LM2596 Buck	DC Barrel (5.5x2.1mm)	2A	Unitree L1 LiDAR

		Converter/ PDB			
P-5V	5.0 VDC	Jetson USB Ports	USB Type-A	3A (Total)	OAK-D Cameras, Wi-Fi Card, Peripherals

5.2 Mechanical Mounting

Component	Dimensions (LxWxH)	Constraint / Interface
Jetson Orin Nano	100mm x 79mm x 21mm	Thermal: Requires 20mm vertical clearance for airflow.
Unitree L1 LiDAR	75mm x 75mm x 65mm	Requires 360° field of view (mounted on mast).
E-Stop	40mm Diameter Head	Must be mounted on the rear exterior panel, accessible without tools.
Motor Drivers (x7 - x9)	Distributed	Thermal: Must be mounted to aluminum chassis for heat dissipation.

5.3 Safety Interlocks

Defines the hardware states that determine if the robot is Safe or Active

State 0: Safe (Passive)

Trigger: E-STOP Button Pressed OR Battery Disconnected.

Hardware State: E-STOP pressed => PDB = 0V.

Result: 22.2V Rail is physically disconnected from Motor Drivers. Motion is impossible.

State 1: Active

Trigger: E-STOP Released AND Master Switch ON.

Hardware State: E-Stop not pressed => PDB = 22.2V.

Result: Motor Drivers energized. Robot responds to commands.

6 Power, mass and financial Budget

Part	Mass	Power	Cost
Lidar L-1	230g	6W	£182
Oak-D Pro Camera	91 g	7.5W	£292 (free)
Oak-D Camera	115 g	5W	£219 (discount: £73)
Power Domain Batt	1357g	-	£196
Coms Domain Batt	297g	-	£30.52
Gigabit Travel Router	181 g	6.5W	£51.75
2x PDB	340.19 g	-	£40.49 + £10
Motor Drives	~1000g	-	£90 - £250
Jetson Orin Nano	176 g	25W	£259
E-Stop	95g	-	£19
Fuses and Switches	10g	-	£10
Power Meter	134 g	-	£10
Total	3.9 kg	50W	£1054 - £1204 (Software Support £350) = £704 - £854

7 Verification plan

7.1 Verification Strategy

The verification process ensures that the as-built rover meets all design requirements defined in the PDR. Each requirement is assigned a verification method:

- **Analysis (A):** Verification via calculation, simulation, or datasheet review.

- **Inspection (I):** Physical examination of the hardware or code.
- **Test (T):** Quantitative measurement of performance under load.

ID	Requirement Description	Method	Pass Criteria	Target Date
SAF-01	Emergency Stop Function	A, T	Pressing the E-Stop button while motors are at 100% throttle must halt all motion within 0.5 seconds.	CDR
SAF-02	Compute Power Isolation	A, I	Activating E-Stop must cut motor power (0V at drivers) while Jetson remains powered and logging.	TRR
POW-01	Battery Runtime	T, A	The 4S Compute Battery must power the Jetson + LiDAR + Cameras for at least 60 minutes under full load. Calculate the Formula for the battery life	TRR
POW-02	Motor Stall Protection	A, I	Research on maximum amperage that the motors can handle and provide a motor controller that's within the parameters.	CDR
NAV-01	Bandwidth Limit	T	Network traffic must remain below 4,000 Kbps average during a 10-minute mapping run.	TRR
NAV-02	Obstacle Detection	T	The Unitree L1 LiDAR and Cameras must detect a standard obstacle at 5 meters range.	TRR
MEC-01	Slope Climbing	T	Rover must ascend a 20° slope with loose sand simulation without back-sliding.	TRR
ENV-01	Thermal Management	T	Jetson CPU temperature must stay below 80°C during 20 minutes of SLAM processing.	TRR

By the Critical Design Review (CDR), the following artifacts will be produced to prove these requirements:

1. A CSV file from the Jetson showing voltage/current draw over a 60-minute test.
2. A screenshot of network load traffic analysis during a teleoperation test.

References

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